A Multivariate Linear Model for Agricultural Data of Iowa for Corn, Soybeans, and Wheat

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Abstract. This paper investigates the joint relationship between Corn, Soybeans, and wheat yield in Iowa, particularly with climatic variables such as precipitation, temperature and anthropogenic factors like nitrogen (fertilizer) use, to fill the existing univariate gap. We took data from the USDA NASS and other sources, primarily from 2020 to 2025. Observing the covariance between the yields of these crops in relation to these variables, also addressing missing data. With this data, we created a multivariate linear model to understand the covariance between these variables and make predictions. The model follows the form $Y=X\beta+\varepsilon$ (or Y=XB+E), where X is the multi-crop yield matrix, β (B) is the coefficient and ε (E) the error terms associated.

1.Introduction

The behavior of agricultural data is theorized to be influenced by environmental factors. To understand the utilization of Iowa and these crops in our model, it is crucial to address the importance of the state of Iowa as an agricultural powerhouse and its top two crops of Corn and Soybeans. This "powerful tag team", started off as horse forage crop, overtime they became the predominant grain crop of Iowa, because of various historical and biological reasons (Birchmier K, 2026). They're agronomically synergistic; soybeans are classified as legumes, they possess the natural ability to improve nitrogen in the soil, because of this, corn production requires less nitrogen fertilization, reducing costs for farmers. The rotation of these two crops on a farm provides pest and disease control, resulting in better health for crops and higher yields. There is also residual management: soybeans are highly adaptable; they can thrive regardless of heavy corn residue. In terms of production, they mostly use the same machinery, and there's a great demand for both (Birchmier, 2026). As for wheat, there is little to no production leading up to and during the 2020-2025 period. We analyzed the yield decay using data from the six-year period between 2013 and 2018, immediately preceding the cutoff. To address this gap, this paper employs alternative methods (discussed subsequently) to substitute for the missing data. For our analysis, we created a multivariate linear model (linear regression).

A multivariate linear model is a statistical method used to analyze the covariance between multiple dependent variables (outcomes) concurrently as a linear combination of multiple predictor variables (independent variables). It models a vector of responses with the goal of understanding how various factors impact multiple outcomes simultaneously. To find the line of best fit (hyperplane) or the coefficient (B) for our model, we used the *ordinary least square method*, which follows the form, $\mathbf{B=(X^TX)^{-1}X^TY}$, where X is the Desing Matrix, which holds the input data (the multi-crop yield matrix); it is an *n x (p + 1)* matrix, consisting of *n* observations of *p* the explanatory (independent) variables, with an added leading column of ones (hence the *p +1*) which represents the existence of an intersection, $\mathbf{X^T}$ is the transpose of **X,** $\mathbf{(X^TX)}$ is the multiplication of the transpose by the original matrix, $\mathbf{(X^TX)^{-1}}$ is the inverse of the product, then it is multiplied by $\mathbf{X^T}$, and finally multiplied by **y**, which is the matrix (or vector) of dependent variables (the revenue). Once these operations are performed, it yields the following:

$$B=\begin{pmatrix}\alpha_0\\ \alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix}$$

Which is the coefficient, it will be in the form $\beta=\alpha_0+\alpha_1X_1+\alpha_2X_2+\alpha_3X_3$, where α_0 is the point of intersection.

2. Preliminary Results

I. Coefficient Operations or Computation[1]

X: the Design matrix or the Multi-Crop Yield Matrix, the columns being Corn, Soybeans, and Wheat respectively, the first column representing intersection.

$$\begin{pmatrix} 1 & 49.2 & 28.92 & 131 \\ 1 & 50 & 31.06 & 146 \\ 1 & 47.5 & 27.25 & 124 \\ 1 & 50.6 & 26.82 & 134 \\ 1 & 51.5 & 36.95 & 140 \\ 1 & 49.7 & 33.59 & 172 \end{pmatrix}$$

X^T: The transpose of the design matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 49.2 & 50 & 47.5 & 50.6 & 51.5 & 49.7 \\ 28.92 & 31.06 & 27.25 & 26.82 & 36.95 & 33.59 \\ 131 & 146 & 124 & 134 & 140 & 172 \end{pmatrix}$$

Price matrix: Contains the prices of the respective commodities from 2020-2025 by the columns[2]

$$\begin{pmatrix} \$3.81 & \$11.1 & \$5.05 \\ \$5.9 & \$13.1 & \$7.63 \\ \$6.4 & \$14.1 & \$8.83 \\ \$5.5 & \$12 & \$6.96 \\ \$5.5 & \$10 & \$6.72 \\ \$4.3 & \$10.5 & \$5.1 \end{pmatrix}$$

Y: The revenue vector, calculated by taking the dot product of each row of both the multi-yield crop matrix (except for the intersection) and the price matrix, yielding a vector.

$$\begin{pmatrix} 1369.72 \\ 2402.72 \\ 2564.01 \\ 2217.46 \end{pmatrix}$$

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[1] Excel & A TI-84 calculator were used for these computations.

[2] The prices were sourced from Statistica

X^TX: Multiplying the transpose of the design matrix by the matrix itself

$$\begin{pmatrix} 6 & 298.5 & 184.59 & 847 \\ 298.5 & 14859.59 & 9199.679 & 42174 \\ 184.59 & 9199.679 & 5756.556 & 26246.64 \\ 847 & 42174 & 26246.64 & 120993 \end{pmatrix}$$

$(X^TX)^{-1}$: Computing the inverse

$$\begin{pmatrix} 345.8576 & -7.55298 & 1.43352 & -0.0994 \\ -7.55298 & 0.17343 & -0.03818 & 0.000705 \\ 1.43352 & -0.03818 & 0.027396 & -0.00267 \\ -0.0994 & 0.000705 & -0.00267 & 0.001038 \end{pmatrix}$$

$(X^TX)^{-1}X^T$: The inverse multiplied by the transpose

$$\begin{pmatrix} 2.68655 & -1.77913 & 13.82846 & -11.1962 \\ -0.03211 & 0.0355 & -0.26811 & 0.292985 \\ -0.0024 & -0.01437 & 0.035443 & -0.12139 \\ -0.00601 & 0.004404 & -0.01001 & 0.003695 \end{pmatrix}$$

$(X^TX)^{-1}X^Ty$: Now multiplying by the revenue matrix (vector)

$$\begin{pmatrix} 3612.275 \\ -19.1548 \\ 9.985724 \\ -6.06813 \end{pmatrix}$$

Our coefficient is **$B=3{,}612.275-19.1548X_1+9.985724X_2-6.06813X_3$**, where **3,612.275** is our point of intersection, the variables X_1, X_2 and X_3 are precipitation, temperature and nitrogen respectively.

II. Computing Error

The last component of a multivariate linear model is e_i, the error or residual[3]. In linear regression models, to compute errors, we must first make predictions. We used the environmental predictors (precipitation, temperature, and nitrogen) respectively.

$$\begin{pmatrix} 49.2 & 28.92 & 131 \\ 50 & 31.06 & 146 \\ 47.5 & 27.25 & 124 \\ 50.6 & 26.82 & 134 \\ 51.5 & 36.95 & 140 \\ 49.7 & 33.59 & 172 \end{pmatrix}$$

[3] Although error and residual are used interchangeably in this paper, it's important to note that, residual is the computation of the difference between the actual and the predicted ($Y-\hat{Y}$), while the error is the theoretical aberration from the actual population model.

To make the predictions, which is denoted as $\hat{\mathbf{Y}}$, we simply substitute the values of the variables (environmental predictors) into the coefficient, and the answer is the prediction or forecast.

Predictions ($\hat{Y}$) of the crop yield (corn, soybeans & wheat) respectively:

$$\begin{pmatrix} 2754.6895 & 2754.6895 & 2754.6895 \\ 2630.6648 & 2630.6648 & 2630.6648 \\ 2812.1792 & 2812.1792 & 2812.1792 \\ 2790.6902 & 2790.6902 & 2790.6902 \\ 2569.2302 & 2569.2302 & 2569.2302 \\ 2421.4359 & 2421.4359 & 2421.4359 \end{pmatrix}$$

The residual (e_i) is said to be the vertical difference between the actual (Y_i) and the predicted ($\hat{Y}_i$) or $\mathbf{e_i = Y_i - \hat{Y}_i}$.

Error Matrix and the sum by year:

$$\begin{pmatrix} -2577.689481 & -2700.689 & -2735.6895 \\ -2426.66476 & -2567.665 & -2581.6648 \\ -2612.17923 & -2753.679 & -2760.1792 \\ -2589.690231 & -2732.69 & -2727.6902 \\ -2358.230219 & -2509.23 & -2501.2302 \\ -2211.435914 & -2357.936 & -2363.4359 \end{pmatrix} \longrightarrow \begin{pmatrix} -8014.07 \\ -7575.99 \\ -8126.04 \\ -8050.07 \\ -7368.69 \\ -6932.81 \end{pmatrix}$$

IV.The Entire Model

We now have all the components of the model ($Y=X\beta+\varepsilon$) respectivly.

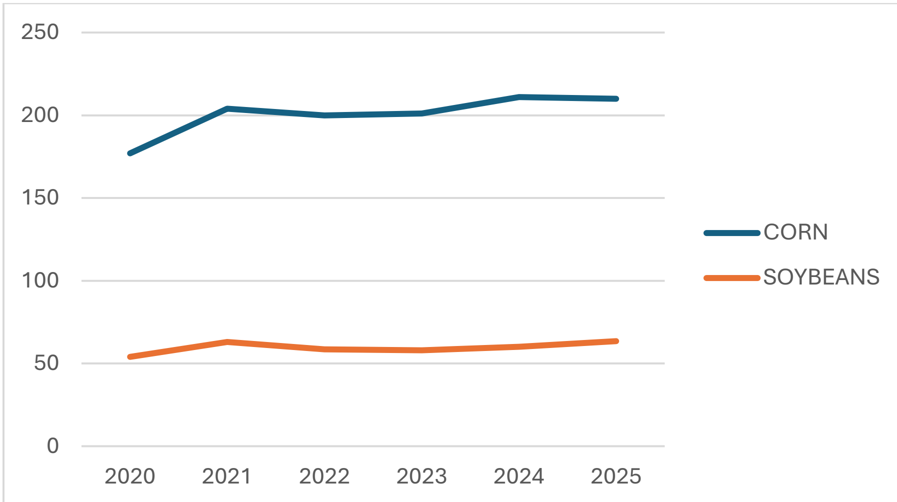
$$\begin{pmatrix} 1369.72 \\ 2402.72 \\ 2564.01 \\ 2217.46 \end{pmatrix} = \begin{pmatrix} 49.2 & 28.92 & 131 \\ 50 & 31.06 & 146 \\ 47.5 & 27.25 & 124 \\ 50.6 & 26.82 & 134 \\ 51.5 & 36.95 & 140 \\ 49.7 & 33.59 & 172 \end{pmatrix} \begin{pmatrix} 3612.275 \\ -19.1548 \\ 9.985724 \\ -6.06813 \end{pmatrix} + \begin{pmatrix} -8014.07 \\ -7575.99 \\ -8126.04 \\ -8050.07 \\ -7368.69 \\ -6932.81 \end{pmatrix}$$

3.Main Problem and Results

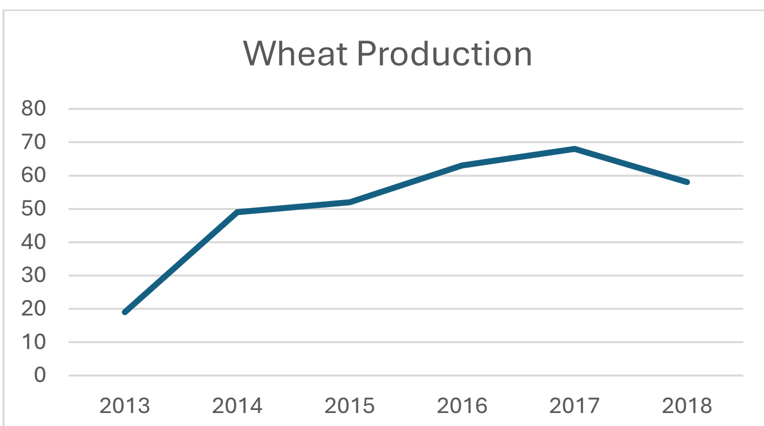
Farmers in Iowa were surveyed in the *Iowa Farm and Rural life Poll* in 2025; the survey reveals the financial obstacles that farmers face which is the cause of this crop production dynamic. (Gordon Arbuckle Jr., 2025) The first was, “Rising input cost and profit margins”, input costs like seed and chemicals tends to increase when “profit margins are higher” 88% of them agreed that these profit margins on corn and soybeans are lost to land rent and input costs, therefore “underscoring persistent financial pressures in row crop production.” Another factor is, Uncertainty and Risk in the economy, 85% of famers simply agreed or strongly agreed that “the boom-bust cycles of commodity production are hard on farmers”, 69% feel sometimes like they have minimal control over profitability, 67% agreed heavy on the financial risks of the heavy reliance on corn and soybeans, and 60% felt “overly dependent on purchased inputs.” . (Gordon Arbuckle Jr., 2025) There are more reasons found in the poll, it’s evident that farmers are not thrilled about their sole reliance on these two commodities. To account for the missing data for wheat, we used the "Regional Representative" Proxy: The Soft Red Winter Wheat (SRW) average for the "Corn Belt" region (US Wheat Associates), it was the closest we could find.

I. Time Series Analysis[4]

We made a time series graph showcasing the yield of corn and soybeans from 2020-2025.



From the graph above it is evident that both crops had a consistent yield over the span of those years, we can assume that farmers were adhering to the quantity of demand. When gathering data for wheat, we saw that the latest data for wheat was from 2018. We decided to see the yield from a six-year (starting from 2013) period until 2018, like we did for the other two commodities.



From 2013 to 2014 we see a steady increase in yield for wheat, then it became slightly constant until 2015, a slight increase till 2017, and from there it started to decrease. These graphs are simple to understand, but we conduct a basic explanation to lay groundwork for further analysis. To continue we've asked various questions, but first is regarding correlation, is there a correlation between the variables (precipitation, temperature and nitrogen) on the yield of these crops.

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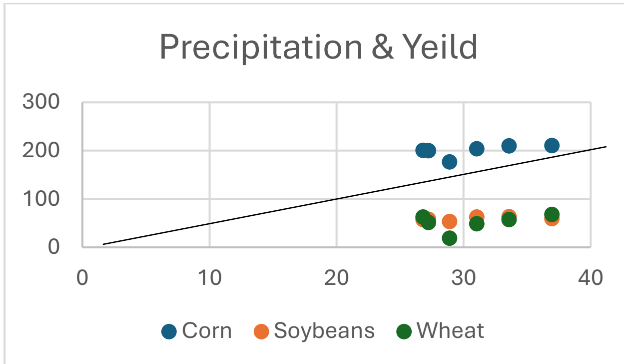
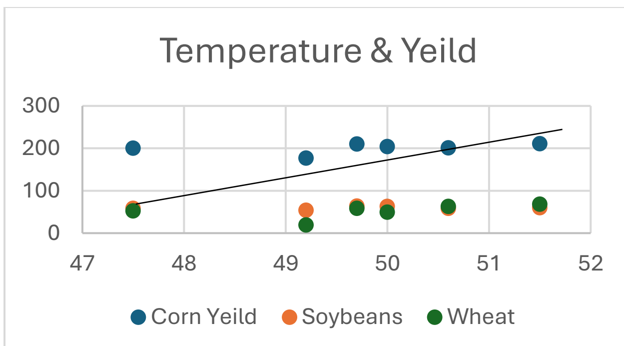
[4] It is important to note that, yes, it is self-evident that there aren't any major discoveries made from these time series graphs, but they're included to provide further context.

II. Correlation

We created a scatter plot to visualize and approximate the correlation (the line) between temperature and yield. From the looks of things, there seems to be some correlation between precipitation and yield. The higher the temperature the more yield we see for corn, and the opposite for soybeans and wheat. We have a *weak positive correlation.*



Now we do the same with temperature and yield, there seems to be a correlation between temperature and yield, the same pattern we see with precipitation. There's a weak positive correlation.



Due to missing data, we couldn't conduct correlation analysis for nitrogen use. It's safe to say that there is a correlation between these variables and crop yield.

4. Conclusion

I. Lurking Variables

Lurking (or confounding) Variables are unaccounted, discreate factors that cause distortion in data between dependent and independent variables, such as false correlations, misguiding and biased outcomes (Frost, 2023). The error we computed in our model reveals that, theoretically, there should be higher yield for these crops in Iowa, in relation to the environmental predictors, but things are more complex. The greatest challenge around lurking variables is how to identify them, because they are unaccounted for.[5] To answer the question, we must "examine data

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[5] "A key question is, 'how does one even identify the existence of a lurking variable when, by definition, it is not among the list of factors contemplated by the analyst?'" Joiner, B. L. (1981). Lurking Variables: Some Examples. *The American Statistician*, *35*(4), 227–233

and residuals with respect to time order”(Joiner, 1981), in our case we found the residual over a six-year period (2020-2025), this is our “diagnoser” variable.[6]

Joiner provided an example, which we found to be the most like our research (model), it’s the “Vitamin B_2 in Turnip Greens”, it consists of a “previously analyzed” data “by both Anderson and Bancroft (1959, p.406) and Draper and Smith (1966, p. 229; 1981, p. 406).”. Similarly, to our paper, they were trying to observe the effects of three variables on a subject (Turnips), them being radiation, soil moisture and temperature in relation to vitamin B in turnips, they “report[ed] a model linear in X_1, X_2, & X_3 [respectively]”. After Draper and Smith computed the residual, they found “runs of + and – signs indicating the presence of unconsidered x-variables” (1966, p. 229; 1981, p. 406)[7].

Although we only saw – signs in the entirety of our residual. The existence of no + signs can also be used to determine the existence of lurking variables. After observing our residual, we concluded that there must be an external factor that we aren’t considering. Then we referred to the *Iowa Farm and Rural life Poll*, we thought that since economic and financial incentives caused decline of wheat yields, we thought that the same factors could also be at play here. It’s important to note that residuals are not always good enough, sometimes the data must be plotted, but in our case it was unnecessary.[8] *Residual matrix below.*

$$\begin{pmatrix} -2577.689481 & -2700.689 & -2735.6895 \\ -2426.66476 & -2567.665 & -2581.6648 \\ -2612.17923 & -2753.679 & -2760.1792 \\ -2589.690231 & -2732.69 & -2727.6902 \\ -2358.230219 & -2509.23 & -2501.2302 \\ -2211.435914 & -2357.936 & -2363.4359 \end{pmatrix} \rightarrow \begin{pmatrix} -8014.07 \\ -7575.99 \\ -8126.04 \\ -8050.07 \\ -7368.69 \\ -6932.81 \end{pmatrix}$$

Economics serve as a lurking variable, because of supply and demand. Theoretically, there can always be more goods being produced, but if demand is not high enough, it is a waste of resources. It’s the most fundamental concept of economics. The financial incentives or concerns are related to the break-even point, this is when revenue equals cost, meaning they made back the amount they spent. They must make more than enough crops, but not too much. Farmers take loans to produce their crops, there’s a lot to lose if they don’t break even, many also have collaterals on their loan, if they can’t pay their debt, they could lose their farms, homes, and livelihoods as they know it, it can take a toll on their mental health, causing them to make a certain irreversible decision; the stakes are too high to lose.

II. Summary of Conclusion

The residual of the model is the basis of our conclusion; we see a constant pattern of negatives. This reveals or causes the speculation of a lurking variable. Theoretically, there should be higher yield due to the environmental predictors, but there isn’t. Lurking or confounding variables are unaccounted factors in mathematical modeling that cause distortion, misguidance, and false correlation in data analysis. The survey *Iowa Farm and Rural life Poll* makes it evident that there are economic and financial explanations such as supply and demand, breaking even, loans and profitability, that affected the yield of corn, soybeans and wheat. Things aren’t always so simple, at times certain elements of influence aren’t so obvious, and majority of the time we have to see the failure of long durations of hard work to come to such realization.[9]

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[6] Although Joiner’s paper focuses on experimental design, it also applies to regression analysis (multivariate)

[7] Joiner referencing Draper and Smith’s textbook “Applied Regression Analysis”. (1966)

[8] “Analyzing the residual does not always make obvious the existence of the lurking variables.” (1981, P.229)

[9] Joiner notes “that lurking variables can be found—or overlooked! —even in such ‘clean’ situations as carefully designed experiments” (1981, p.227)

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